Haonan Chen
1308 W Main St
Urbana, IL 61801
+1 2178191032
haonan2@illinois.edu

CHAI Selection Committee
University of California, Berkeley

March 12, 2025

Dear CHAI Selection Committee,

I am writing to express my interest in collaborating with the Center for Human-Compatible AI (CHAI). My research focuses on developing **deployable and pragmatic robots** that can rapidly acquire reasoning and manipulation capabilities, much like how humans learn to interact with the world. I am eager to contribute to CHAI's mission of advancing provably beneficial AI systems by integrating insights from **human priors**, enabling intuitive physics, predictive modeling, and tool affordance learning.

As a Ph.D. candidate in ECE at UIUC, advised by Professor Katie Driggs-Campbell, my research integrates human priors into robot decision-making, enabling robots to reason, predict, and adapt efficiently. I developed graph-based neural dynamics models to predict and reason about object interactions in cluttered environments (**Best Paper Finalist at CoRL 2023**). My work also enhances coordinated bimanual manipulation by integrating state prediction with inverse dynamics models, enabling robots to anticipate future states and synchronize bimanual actions in tasks such as laundry cleanup, bimanual fruit holding, and cluttered shelf picking. I also developed cross-embodiment tool-use learning from human play data, enabling fast skill acquisition with generalization across viewpoints, embodiments, and objects.

My future research will focus on advancing **composable policy and model learning** to enable robots to flexibly generalize across tasks, environments, and embodiments. Current robot learning approaches falter due to rigid, data-hungry paradigms. To address this, I propose a structured hybrid learning framework that dynamically integrates model-based reasoning and policy-based learning, allowing robots to switch adaptively between structured world models and direct policy learning based on function complexity. My approach emphasizes modular and composable learning, where object dynamics, affordances, and skill representations remain independent yet seamlessly combinable. By unifying learned models and policies, robots can rapidly adapt to new environments with minimal fine-tuning, significantly reducing reliance on large-scale retraining.

Beyond research, I am passionate about mentorship and fostering the next generation of AI and robotics researchers. At UIUC, I served as a TA for ECE 484 (Safe Autonomy) with Professor Sayan Mitra, where I developed a lane detection machine problem to introduce students to real-world perception challenges in autonomous systems in Spring 2025. Beyond the classroom, I have mentored seven undergraduate students and one master student, guiding them through research in robotic manipulation and machine learning. I am committed to mentoring students at all levels, fostering their engagement in independent research and problem-solving.

I am excited about the prospect of working with CHAI and would welcome the opportunity to discuss how my expertise can contribute to its research initiatives. Thank you for your time and consideration.

Sincerely,



Haonan Chen